

A Tensorization Obstruction to a Scalar Closure-Defect Repair for a Union-Closed Entropy Functional

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Abstract

We analyze a scalar repair proposed for a Shapley-averaged entropy functional arising in an approach to the union-closed sets conjecture. For a finite set family $\mathcal{F}$, let $A_+(\mathcal{F})$ combine the iid OR-entropy and an optimized one-sided causal Bellman coupling, and let $\varepsilon_V(\mathcal{F})$ be the probability that the union of two independent uniform members leaves $\mathcal{F}$. The question was whether $-A_+(\mathcal{F})/\varepsilon_V(\mathcal{F})$ is universally bounded over cap-2/5, Reimer-admissible families. We prove that it is not. A 45-member normalized family on seven coordinates has a certified value $A_+ < -7/250$. Both iid and optimized causal costs are additive under Cartesian products, whereas closure success is multiplicative. Its k -fold powers are normalized and preserve the cap and Reimer conditions, but their repair ratios exceed $7k/250$. A standalone 256-bit Arb verifier certifies the sharper base value, directly checks the 2,025-member square, and evaluates the actual square Bellman recursion for two contrasting global orders. Independently, a standard-library 160-bit dyadic checker relaxes every nondegenerate action interval to $[0, 1]$ and still proves $A_+ < -1/40$, without clamp decisions or Arb. The same dyadic primitive certifies a separate 25-member family on six coordinates, evaluating all 720 orders without a symmetry quotient and proving $A_+ < -1/80$. Its Cartesian powers have repair ratio greater than $k/80$. Finally, an exact rational certificate evaluates the true optimum over the exact feasible action set, with no interval library, no clamp classification and no relaxation, and returns two-sided enclosures of width below 6×10^{-27} for both bases over all 720 and all 5040 coordinate orders. These constructions rule out every fixed scalar multiple of this ordered-pair defect. The numerator's growth is then closed on both sides: incidence is at most $n \lfloor 2m/5 \rfloor$ while Reimer demands at least $m \log_2 m/2$, so every admissible family obeys the size ceiling $\log_2 m \leq 4n/5$, and writing $\Lambda(n)$ for the supremum of $-A_+$ over families on at most n coordinates, $\frac{3}{640}n - \frac{3}{80} \leq \Lambda(n) \leq \frac{4}{5}n$. The lower constant uses a cloned-coordinate base, admissible but not separating, which improves the published slope $1/250$ by exactly $75/64$; no step of the product argument uses separation, and Reimer for every power reduces to the single base inequality $m^m \leq 2^{2I}$. Because closure success multiplies under products, no Cartesian power reaches small defect, so the small-defect regime needs separate evidence. We prove two defect floors — one from the cap, one from downset counting, the second attained — which together show the repair ratio is $O(n)$ on every family without a set of size at least $\frac{8}{5}\bar{s} \geq \frac{4}{5}\log_2 m$. Exact direct witnesses push the least certified negative defect to 1144/1875 among separating families and 4/9 without separation, with best ratios 531/11360 and 147/2536 respectively. We then prove a sharp clone-order theorem: clone multiplicities induce the exact Plackett–Luce law on original fixed orders, and their infimum is the minimum fixed-order objective. A row-changing search finds a normalized eight-coordinate core at defect 14/45 with a negative fixed order; exact reweighting yields a finite non-separating negative witness at the same defect, crossing 2/5. A different 14/45 core and all its defect-free extensions remain certified positive, illustrating why the fixed-order criterion is necessary. Finally, the Chase–Lovett approximate-union-closed construction rules out any defect floor from frequency cap and a dominant set alone, but violates Reimer by $(h(\psi)/2 - \psi)n > 0.0977n$. Thus a Gate B local floor, if true, must use Reimer essentially. None of these statements produces a union-closed counterexample or settles the remaining Reimer-admissible local question.

1 Problem and definitions

Let $\mathcal{F} \subseteq 2^{[n]}$ be a nonempty finite simple family, $m = |\mathcal{F}|$, and let all random members be uniform on $\mathcal{F}$. The admissibility conditions are

$$\max_{i \in [n]} |\{A \in \mathcal{F} : i \in A\}| \leq \left\lfloor \frac{2m}{5} \right\rfloor, \quad (1)$$

$$\sum_{A \in \mathcal{F}} |A| \geq \frac{m \log_2 m}{2}. \quad (2)$$

A family is called normalized here when every coordinate column is nonzero and the columns are pairwise distinct. Normalization is not required in the supremum below, but all members of both sequences satisfy it.

Fix a coordinate order π . At coordinate i and predecessor prefix u , write

$$p_i(u) = \mathbb{P}[X_i = 1 \mid X_{\text{pred}_\pi(i)} = u].$$

For two independent prefixes u, v , define

$$Q_\pi(\mathcal{F}) = \sum_i \mathbb{E}_{u,v} h(p_i(u) + p_i(v) - p_i(u)p_i(v)),$$

where h is binary entropy in bits, and put $Q(\mathcal{F}) = \mathbb{E}_\pi Q_\pi(\mathcal{F})$.

For the causal term, at a pair of prefixes let $p = p_i(u)$ and $r = p_i(v)$. Set

$$s^*(p, r) = \max\{p, r, \min(p + r, 1/2)\}, \quad U(p, r) = \min(1, p + r).$$

A one-sided action chooses $s \in [s^*, U]$, with next-bit probabilities

$$P_{00} = 1 - s, \quad P_{10} = s - r, \quad P_{01} = s - p, \quad P_{11} = p + r - s.$$

The fixed-order Bellman value $C_{+,\pi}(\mathcal{F})$ is the maximum expected sum of $h(s)$ over all such causal policies. We average over all orders to obtain $C_+(\mathcal{F})$. The coefficient is the fixed, dimension-independent rational

$$\alpha = 0.0356069 = \frac{356069}{10^7},$$

and the functional is

$$A_+(\mathcal{F}) = (1 - \alpha)Q(\mathcal{F}) + \alpha C_+(\mathcal{F}) - \log_2 m. \quad (3)$$

The ordered-pair closure defect is

$$\varepsilon_\vee(\mathcal{F}) = \mathbb{P}_{X,Y \sim \text{Unif}(\mathcal{F})}[X \cup Y \notin \mathcal{F}]. \quad (4)$$

The Gate B quantity is

$$c_{\text{cl}}^* = \sup_{\substack{\mathcal{F} \text{ satisfies (1), (2)} \\ A_+(\mathcal{F}) < 0}} \frac{-A_+(\mathcal{F})}{\varepsilon_\vee(\mathcal{F})}. \quad (5)$$

The quotient is interpreted in the extended nonnegative reals: an admissible family with $A_+(\mathcal{F}) < 0$ and $\varepsilon_\vee(\mathcal{F}) = 0$ has value $+\infty$; otherwise the quotient is taken only at positive defect. Our construction has strictly positive defect, so the theorem is independent of this convention. No upper bound on $\varepsilon_\vee$ occurs in (5).

2 Certified negative base

Partition the seven coordinates as $\{5, 6\} \sqcup \{0, 1, 2, 3, 4\}$ and let $\mathcal{B}$ be the union of the seven weight cells

$$(0, 1), (0, 3), (1, 0), (1, 2), (1, 5), (2, 0), (2, 1). \quad (6)$$

The exact row-mask list is stored in the accompanying machine-readable candidate. Direct counting gives

$$|\mathcal{B}| = 45, \quad d_i = 18 \quad (0 \leq i < 7), \quad \sum_{A \in \mathcal{B}} |A| = 126.$$

Thus the cap holds with equality. The exact Reimer threshold is 124. The columns are active and pairwise distinct. There are exactly 1,600 missing ordered join pairs, and hence

$$\varepsilon_V(\mathcal{B}) = \frac{64}{81}, \quad 1 - \varepsilon_V(\mathcal{B}) = \frac{17}{81}. \quad (7)$$

The family is not union-closed: rows 1 and 2 occur but row 3 does not.

Proposition 1 (Certified base). *For the functional (3),*

$$A_+(\mathcal{B}) \leq -0.0286491867944683178160269789255044875 \dots < -\frac{7}{250}. \quad (8)$$

The proposition is a finite computational lemma. A standalone checker reconstructs $\mathcal{B}$ from the cell predicate in (6), verifies that it equals the stored row list, rebuilds the exact constraints and the 240-element automorphism group, reduces the 5,040 coordinate orders to 21 equal-size orbits, and produces a rigorous upper certificate for the one-sided Bellman maximum with 256-bit Arb arithmetic. At each of 982 clamp-ambiguous states the unconstrained concave maximum is retained, which upper-bounds the constrained maximum. The checker imports no implementation module from the original discovery code.

Proposition 2 (Independent clamp-free certificate). *There is a standard-library-only, outward-rounded dyadic certificate that proves*

$$A_+(\mathcal{B}) < -0.0271742911034863966862278304907 < -\frac{1}{40}.$$

Proof. For a fixed order, define a relaxed backward value W as follows. At a nondegenerate state, replace the feasible one-sided action interval by all of $[0, 1]$. Given child values W_{ac} , put

$$D = -W_{00} + W_{10} + W_{01} - W_{11}.$$

The unrestricted concave maximum is

$$W_{00} - rW_{10} - pW_{01} + (p + r)W_{11} + \log_2(1 + 2^D).$$

At a degenerate state the action is forced and the boundary recurrence is used. Backward induction gives $C_{+, \pi} \leq W_\pi$: before enlarging the action interval, all four feasible transition coefficients are nonnegative, so replacing each true child by an upper child preserves the inequality; maximizing over a superset can only increase it.

The checker reconstructs $\mathcal{B}$ from (6), computes Q directly by the predecessor-subset Shapley formula, and exactly partitions all 5,040 orders into the 21 equal orbits of the verified 240-element automorphism group. It obtains the outward upper endpoints

$$\begin{aligned} Q(\mathcal{B}) &< 5.447354977380321988160699884419, \\ \mathbb{E}_\pi W_\pi &< 5.933885055704082998831935023544, \end{aligned}$$

which imply (2). Natural logarithms use a rational atanh series with an exact tail bound; exponentials use range reduction, a rational Taylor tail bound, and repeated squaring. Thus no floating-point library or Arb operation enters this second certificate. $\square$

3 Tensorization

For families on disjoint coordinate blocks define

$$\mathcal{F} \boxtimes \mathcal{G} = \{A \cup (n + B) : A \in \mathcal{F}, B \in \mathcal{G}\}.$$

Lemma 3 (Closure success).

$$1 - \varepsilon_V(\mathcal{F} \boxtimes \mathcal{G}) = (1 - \varepsilon_V(\mathcal{F}))(1 - \varepsilon_V(\mathcal{G})). \quad (9)$$

Proof. A product join lies in the product family if and only if both block joins lie in their respective families. Under two independent uniform product rows, the two events are independent. $\square$

Lemma 4 (Cost tensorization).

$$Q(\mathcal{F} \boxtimes \mathcal{G}) = Q(\mathcal{F}) + Q(\mathcal{G}), \quad C_+(\mathcal{F} \boxtimes \mathcal{G}) = C_+(\mathcal{F}) + C_+(\mathcal{G}). \quad (10)$$

Proof. Fix a global coordinate order π and write π_F, π_G for its induced relative block orders. A uniform product row has independent blocks. Conditioning on predecessor coordinates in the other block therefore does not alter a within-block conditional-one probability. Each iid local term is the corresponding factor term, giving

$$Q_\pi(\mathcal{F} \boxtimes \mathcal{G}) = Q_{\pi_F}(\mathcal{F}) + Q_{\pi_G}(\mathcal{G}).$$

For the optimized causal term, split every pair of product prefixes into its two block states. We prove by backward induction on the remaining global suffix that the Bellman value decomposes as $V = V_F + V_G$. The terminal case is immediate. Suppose the next coordinate belongs to the F block. Its conditional marginals, action interval, and transition probabilities depend only on the F prefixes. By induction, every child value is its F -child value plus the same unchanged constant V_G . For every feasible action,

$$\sum_{a,c} P_{ac} = (1 - s) + (s - r) + (s - p) + (p + r - s) = 1,$$

and hence

$$\max_s \left\{ h(s) + \sum_{a,c} P_{ac} (V_{F,ac} + V_G) \right\} = V_G + \max_s \left\{ h(s) + \sum_{a,c} P_{ac} V_{F,ac} \right\}.$$

The common shift also cancels from $-V_{00} + V_{10} + V_{01} - V_{11}$, so it cannot change the optimizer. The remaining maximization is exactly the F Bellman step. The other block is symmetric. In particular, a policy cannot gain by using other-block prefixes: their continuation contribution cancels from the Bellman slope and hence from the optimizer. Thus

$$C_{+,\pi}(\mathcal{F} \boxtimes \mathcal{G}) = C_{+,\pi_F}(\mathcal{F}) + C_{+,\pi_G}(\mathcal{G}).$$

A uniform global order induces uniform relative orders in each block. Averaging proves (10). $\square$

Corollary 5.

$$A_+(\mathcal{F} \boxtimes \mathcal{G}) = A_+(\mathcal{F}) + A_+(\mathcal{G}). \quad (11)$$

Proof. Combine (10) with $\log_2 |\mathcal{F} \boxtimes \mathcal{G}| = \log_2 |\mathcal{F}| + \log_2 |\mathcal{G}|$ in (3). $\square$

4 Unbounded repair ratio

Let $\mathcal{F}_k = \mathcal{B}^{\boxtimes k}$. Then $|\mathcal{F}_k| = 45^k$ and the dimension is $7k$. Every coordinate occurs in

$$18 \cdot 45^{k-1} = \frac{2}{5} 45^k$$

rows, so the cap holds with equality. The total incidence is $126k45^{k-1}$ and the average row size is $14k/5$. Since

$$45^5 = 184,528,125 < 268,435,456 = 2^{28},$$

we have $\log_2 45 < 28/5$, and (2) holds strictly for every k . Activity and separation within a block are inherited from $\mathcal{B}$; columns in different blocks are distinct because base columns take both values and blocks vary independently. Hence every $\mathcal{F}_k$ is normalized.

Equations (7), (9), and (11) give

$$\varepsilon_V(\mathcal{F}_k) = 1 - \left(\frac{17}{81}\right)^k, \quad A_+(\mathcal{F}_k) = kA_+(\mathcal{B}).$$

Theorem 6 (Gate B). *The quantity in (5) is infinite.*

Proof. By (8),

$$\frac{-A_+(\mathcal{F}_k)}{\varepsilon_V(\mathcal{F}_k)} > \frac{7k/250}{1 - (17/81)^k} > \frac{7k}{250} \rightarrow +\infty.$$

Every member of the sequence is admissible, so the supremum is infinite. $\square$

4.1 A second base

The conclusion does not rely on the seven-coordinate search output. Partition six coordinates into two triples and let $\mathcal{D}$ be the union of the weight cells

$$(0, 0), (0, 1), (1, 0), (1, 2), (2, 1).$$

It has 25 rows, every coordinate count is 10, total incidence 60, and exact Reimer threshold 59. Exactly 444 of its 625 ordered joins are missing, so its closure-success probability is $181/625$. A third-party-free 160-bit dyadic checker evaluates the clamp-free relaxation for all $6! = 720$ orders, without an automorphism quotient, and certifies

$$\begin{aligned} Q(\mathcal{D}) &< 4.616383859857110260038923460896, \\ \mathbb{E}_\pi W_\pi(\mathcal{D}) &< 5.034353951629763221689815640761, \\ A_+(\mathcal{D}) &< -0.0125897106568747589615082453240 < -\frac{1}{80}. \end{aligned}$$

A separate 256-bit Arb evaluation of the same family and its 10 exact order orbits certifies the stronger bound

$$A_+(\mathcal{D}) < -\frac{17}{1250}.$$

Thus the second base, like the first, survives both arithmetic formulations; the dyadic certificate additionally removes the order quotient. For $\mathcal{G}_k = \mathcal{D}^{\boxtimes k}$, the product identities give

$$A_+(\mathcal{G}_k) = kA_+(\mathcal{D}), \quad \varepsilon_V(\mathcal{G}_k) = 1 - \left(\frac{181}{625}\right)^k.$$

The cap holds with equality. The average row size is $12k/5$, and $25^5 < 2^{24}$ proves the strict Reimer inequality. Hence this second sequence is normalized and admissible and independently gives

$$\frac{-A_+(\mathcal{G}_k)}{\varepsilon_V(\mathcal{G}_k)} > \frac{k/80}{1 - (181/625)^k} > \frac{k}{80} \rightarrow +\infty.$$

4.2 Exact rational certificate without relaxation

The two certificates above are numerical in different ways: one needs a third-party ball-arithmetic library and a three-way clamp classification, the other enlarges every feasible action interval to $[0, 1]$ and therefore bounds a relaxation of C_+ rather than C_+ itself. A third certificate removes both concessions.

Proposition 7 (Exact rational enclosures). *There are exact rational two-sided enclosures, computed with ordinary fractions and no third-party package, of the true objective over the exact feasible action set:*

$$A_+(\mathcal{D}) \in [-0.0136721077321777735618599156610, -0.0136721077321777735618599129978], \quad (12)$$

$$A_+(\mathcal{B}) \in [-0.0286491867944683178160269819327, -0.0286491867944683178160269764548]. \quad (13)$$

Both widths are below 6×10^{-27} , and the upper endpoints give $A_+(\mathcal{D}) < -17/1250$ and $A_+(\mathcal{B}) < -7/250$.

Proof. Three certified scalar primitives suffice, all integer arithmetic. First, for $y > 0$ and any bit b one has $\log_2 y = (b + \log_2(y^2/2^b))/2$ exactly; iterating N times with monotone rounding of the state keeps the accumulated value on the intended side, and choosing $b = 1$ exactly when the rounded square is at least two keeps the state in $(0, 2]$, so the discarded tail $2^{-N} \log_2 y_N$ lies in $[0, 2^{-N}]$. Second, with $r_0 = 2$ and r_{j+1} the ceiling square root of r_j at a fixed dyadic denominator, induction gives $r_j \geq 2^{2^{-j}}$, so rounding an exponent up to N binary places and multiplying the r_j over its set bits bounds 2^x from above. Third, both weights of h are negative, so bounding each logarithm on the opposite side bounds h ; and $h(s) + cs$ is concave with derivative $\log_2((1-s)/s) + c$, so on $[\ell, u]$ its maximum is exactly at an endpoint when the derivative has a sign there and otherwise at most the Fenchel value $\log_2(1 + 2^c)$.

A Bellman step is monotone in its children because the four feasible transition probabilities are nonnegative and sum to one; replacing children, and then the affine data, by upper bounds therefore yields an upper bound. For the lower endpoint the same recursion evaluates one explicitly feasible rational action near $1/(1 + 2^{-c})$ against pessimistic children, which is the value of an admissible one-sided coupling and hence at most $C_{+, \pi}$. Evaluating all 720 and all 5040 coordinate orders — without any automorphism quotient — and averaging gives the stated enclosures; the all-order value lists partition into 10 classes of size 72 and 21 classes of size 240, which recovers the automorphism data instead of assuming it. $\square$

The Arb value in (8) and its six-coordinate analogue lie strictly inside these enclosures, so retaining the unconstrained maximum at the 982 and 305 clamp-ambiguous states cost less than 10^{-26} .

4.3 Growth rate

Let $c^*(n)$ denote the supremum in (5) restricted to admissible families on at most n coordinates. Taking $k = \lfloor n/d \rfloor$ powers of a certified base of dimension d with $A_+ \leq -\delta$ gives

$$c^*(n) \geq \frac{\delta \lfloor n/d \rfloor}{1 - (1 - \varepsilon_V)^{\lfloor n/d \rfloor}} > \delta \left\lfloor \frac{n}{d} \right\rfloor,$$

so $c^*(n) > n/250 - 7/250$ from $\mathcal{B}$ and $c^*(n) > 17n/7500 - 17/1250$ from $\mathcal{D}$. In the other direction every local term of Q is a binary entropy and C_+ is a maximum of sums of binary entropies, so $Q \geq 0$ and $C_+ \geq 0$, whence $A_+ \geq -\log_2 m \geq -n$. Therefore any admissible family on at most n coordinates with $\varepsilon_V \geq \varepsilon_0 > 0$ satisfies $-A_+/\varepsilon_V \leq n/\varepsilon_0$. Under a positive defect floor the Gate B ratio is thus of order exactly n , and the only route to a bounded ratio is to force $\varepsilon_V \rightarrow 0$.

5 Computational controls

The proof uses computation only to establish the two strictly negative finite bases. The standalone verifiers additionally perform the following adversarial checks.

1. It explicitly constructs the 2,025-row square and counts 3,920,000 missing ordered joins, obtaining $\varepsilon_{\vee}(\mathcal{F}_2) = 6272/6561 = 1 - (17/81)^2$.
2. It evaluates the actual square-family Bellman recursion in an exact factored-fiber representation for consecutive and alternating global orders. Both Arb enclosures agree with twice the base-order optimum, with residual radius below 1.4×10^{-70} . The alternating computation visits 2,985,831 states and encounters 405,641 ambiguous clamps, retaining the unconstrained upper bound at each.
3. It records $\alpha = 356069/10^7$ independently of dimension and checks the cap, incidence, Reimer witness, and exact defect arithmetic for powers $1 \leq k \leq 8$. The all- k conclusion is the proof above, not extrapolation from these instances.
4. A separate regression suite attacks the fixed-order product identity for all 225 ordered pairs of nonempty two-coordinate families and all 24 global orders of each product: 5,400 interleaved Bellman comparisons. It checks the iid and one-sided identities with the legacy float implementation and the defect identity exactly. It also compares the independent Arb and legacy float Bellman evaluators and brute-checks the $n = 8$ order quotient.
5. A non-factored audit materializes the same square as a plain list of 2,025 fourteen-bit masks and runs generic prefix-fiber evaluators on it, with no product-aware shortcut, for five declared global orders: consecutive, block-swapped, reversed, alternating, and a seeded interleaving. The directly computed $C_{+,\pi}$ and Q_{π} match the induced block sums to float64 rounding, worst gaps 8.9×10^{-16} and 3.6×10^{-15} , while the values take three distinct magnitudes.
6. The second base has its own non-factored control. It materializes the 625-row square as plain 12-bit masks, directly counts 357,864 missing ordered joins out of 390,625, and runs the generic evaluators for the same five order patterns. The Bellman product gap is zero in every case and the worst iid gap is 1.8×10^{-15} . This resumable check does not use the dyadic certificate's order computation.
7. A resumable falsification search attacks additivity on 292 ordered products, 205 with unequal block dimensions, including dead columns, duplicate columns, singletons, full power sets, middle layers, and seeded random pairs, over every global order of each product. It found no counterexample; the worst gap is 1.8×10^{-15} and the exact ordered-defect product never failed.
8. The same attack is then repeated in exact rational arithmetic on 35 ordered products, 20 with unequal block dimensions, over every global order of every product, 15,010 orders in total. For each order it requires the certified enclosures of the two sides to intersect, so a failure would be a counterexample and not rounding. There were no intersection failures, no failures of the exact defect identity, a worst Q discrepancy of 1.4×10^{-27} , and a worst fixed-order Bellman discrepancy of exactly zero.
9. The shared algebra is verified symbolically: the four transition weights sum identically to one, the step objective equals $\text{const} + sD + h(s)$ so a common additive shift of the continuation values leaves D unchanged, and $\max_s \{sD + h(s)\} = \log_2(1 + 2^D)$ at $s = 1/(1 + 2^{-D})$ with second derivative $1/(s(s-1)\ln 2) < 0$.

10. The powers themselves are instantiated for $k \leq 3$ and re-audited from the constructed rows. At $k = 3$ the family has 91,125 members, all 21 coordinate counts equal the cap bound 36,450, incidence 765,450 exceeds the Reimer threshold 750,668, the columns are active and pairwise distinct, and a brute-force count over all 8,303,765,625 ordered pairs finds 8,227,000,000 missing joins, i.e. exactly $526528/531441 = 1 - (17/81)^3$. That count uses no product shortcut.

Complete smaller-dimension controls remain separate from the theorem. The full labeled $n = 4$ scan (which also contains lower-dimensional families via zero columns) retained 358 nontrivial cap-2/5 families. An exact $n = 5$ producer represented all 463,343 nontrivial admissible labeled families by 5,172 coordinate orbits; a resumable 120-order evaluation found minimum $A_+ = 0.006474313148642441 \dots$. At $n = 6$, exhaustive coverage is feasible only for declared subclasses: all 262,585 admissible families with $3 \leq |\mathcal{F}| \leq 7$ (1,081 orbits), and all 450 orbits in the three two-block classes. Enumeration and defects in these controls are exact; their objective comparisons are float64. None is used to infer the all- k result.

A complete declared-class $n = 8$ search provides a finite control. It exhausts all 65,535 families invariant under $S_1 \times S_7$, finds 136 admissible full S_8 orbits (130 normalized), and finds two negative values. Its largest float64 ratio is $0.034335490331443 \dots$, below the certified seven-coordinate base. This is a complete numerical result for that symmetry class, not an arbitrary- $n = 8$ optimum.

6 The size ceiling and the local frontier

Lemma 8 (size ceiling). *Every admissible $\mathcal{F} \subseteq 2^{[n]}$ satisfies $\log_2 m \leq 4n/5$, equivalently the integer inequality $m^5 \leq 2^{4n}$.*

Proof. Incidence counted by coordinates is $\sum_{A \in \mathcal{F}} |A| = \sum_{i \in [n]} \deg_i$, and the cap bounds each degree by $\lfloor 2m/5 \rfloor \leq 2m/5$, so the incidence is at most $2nm/5$. Reimer admissibility demands incidence at least $m \log_2 m/2$. Divide $m \log_2 m/2 \leq 2nm/5$ by $m > 0$. $\square$

With $Q, C_+ \geq 0$ this gives $-A_+ \leq \log_2 m \leq 4n/5$ unconditionally, so together with the powers the supremum $\Lambda(n)$ of $-A_+$ over families on at most n coordinates satisfies $(7/250)\lfloor n/7 \rfloor \leq \Lambda(n) \leq 4n/5$: the numerator's growth is exactly of order n . The bound is close to sharp; the largest admissible sizes are 45 at $n = 7$ and 445 at $n = 11$, against $2^{5.6}$ and $2^{8.8}$.

Lemma 9 (defect floors). *Write $M = \max_{A \in \mathcal{F}} |A|$, $\bar{s} = \frac{1}{m} \sum_A |A|$ and $N(A) = |\{B \in \mathcal{F} : B \subseteq A\}|$. Then, if $M < n$,*

$$\varepsilon_V \geq \frac{\frac{8}{5}\bar{s} - M}{n - M}, \quad \text{and always} \quad \varepsilon_V \geq 1 - \frac{1}{m^2} \sum_{A \in \mathcal{F}} N(A)^2.$$

Proof. With $p_i = \deg_i/m \leq 2/5$, $\mathbb{E}|X \cup Y| = \sum_i p_i(2 - p_i) \geq \frac{8}{5}\bar{s}$, while a union lying in $\mathcal{F}$ has size at most M and any union has size at most n , so $\mathbb{E}|X \cup Y| \leq M + \varepsilon_V(n - M)$. For the second bound, $X \cup Y = A$ forces $X, Y \subseteq A$ with $X, Y \in \mathcal{F}$, so at most $N(A)^2$ ordered pairs have union A ; summing over $A \in \mathcal{F}$ bounds $(1 - \varepsilon_V)m^2$. $\square$

A third base. A defect-minimizing search at $n = 7, m = 45$ produced a normalized family $\mathcal{L}$ (machine-readable rows in `certificates/gate_b_lowdefect_rational_v1.json`) with every degree at the cap 18, incidence $126 \geq R_{45} = 124$, and defect $1336/2025$. Its automorphism group is trivial, so no quotient is available and all 5040 fixed-order enclosures are distinct; the exact rational certificate proves $A_+(\mathcal{L}) < -1/1200$, hence base ratio $-A_+/\varepsilon_V > 27/21376$. Its powers satisfy $\varepsilon_V(\mathcal{L}^{\boxtimes k}) = 1 - (689/2025)^k$ and $A_+(\mathcal{L}^{\boxtimes k}) = kA_+(\mathcal{L})$, giving the third explicit diverging

sequence. It is not needed for the main theorem, but makes the local-frontier and “all three bases” statements below self-contained.

The second bound is attained: over all 366 admissible families with $n \leq 4$ its tightest slack is exactly 0. Combining the first with Lemma 8, if $M \leq \theta \cdot \frac{8}{5}\bar{s}$ with $\theta < 1$ then $-A_+/\varepsilon_v \leq 5n/(4(1-\theta))$. Hence the repair ratio is $O(n)$ on every family without a dominant set, and any approach to the local regime requires a set of size at least $\frac{8}{5}\bar{s} \geq \frac{4}{5}\log_2 m$. Both seven-coordinate bases have such a set, and every eight-coordinate witness below contains [8].

A further consequence is that 25 and 45 are the largest admissible sizes at $n = 6, 7$; all three bases of at most seven coordinates attain them with every degree exactly at the cap, and no set whatsoever can be added to any of them. Lowering the defect therefore requires a larger ground set.

Accordingly the complete $S_2 \times S_6$ class at $n = 8$ was enumerated: 2,272 canonical admissible families, of which 21 have negative float64 objective. Exact rational re-evaluation of all 36 candidates with float value below 0.01 certifies 20 of them and shows one, at defect 2553/3200, is exactly *positive*; no family that reads non-negative in float64 is exactly negative. The discrepancy is not rounding: on a fixed order set the float and exact evaluators agree to 10^{-14} , and the exact orbit route reproduces the average over all $8!$ orders exactly. The census averages $C_{+,\pi}$ over one order per $S_k \times S_{8-k}$ pattern, which represents the objective only when the family’s automorphism group *is* that block group. Here it is not: $|\text{Aut}| = 1440$ coincides with $|S_2 \times S_6|$, yet only 120 of those elements preserve the block partition, so the 28 block-pattern orders cover only 12 of the 28 true orbits, six of them repeatedly. The induced error in A_+ reaches 2.03×10^{-3} , and any subset average is within $\alpha(\max_\pi C_{+,\pi} - \min_\pi C_{+,\pi}) \leq 5.2 \times 10^{-3}$ of the truth, which is why candidates are re-ranked exactly above a threshold of +0.01 rather than 0. The certified witnesses give

$$\min\{\varepsilon_v : A_+ < 0\} \leq \frac{1144}{1875} = 0.6101\dots \quad (\text{separating}), \quad \leq \frac{139}{245} = 0.5673\dots,$$

improving 1336/2025, and a separating 75-row family with $\varepsilon_v = 468/625$ has certified ratio at least $77/2080 = 0.0370\dots$, the first here above the base ratio $0.0363\dots$. Sizes 70 and 75 are infeasible at $n = 7$, since $7\lfloor 2 \cdot 70/5 \rfloor = 196$ is below the Reimer threshold 215; the eighth coordinate supplies the missing incidence, and in the 70-row families it duplicates the first coordinate exactly. Coordinate cloning changes neither the defect nor the size, so it is precisely the mechanism Lemma 8 predicts. These two certificates use the exact symmetry quotient ($|\text{Aut}| = 1440$, 28 orbits of size 1440), and both facts that make a uniform average over representatives equal to the average over all $8!$ orders are tested rather than assumed.

6.1 Separation is not used, and the growth constant improves

No step of Section 8’s product argument uses separation. Lemma 1 uses only that the two block join events are independent under two independent uniform product rows; Lemmas 2 and 3 use only that a uniform product row has independent blocks, that the conditional marginals, feasible interval and four transition probabilities at a coordinate depend on that block’s prefixes alone, and that the four transition probabilities sum to one; additivity of A_+ uses only $|\mathcal{F} \boxtimes \mathcal{G}| = |\mathcal{F}||\mathcal{G}|$. Separation appears once, where it is *concluded* for the powers of a normalized base, never assumed.

Admissibility of the powers needs nothing beyond the base. For a base with size m , incidence I and every degree equal to c , the power $\mathcal{F}^{\boxtimes k}$ has size m^k , degree cm^{k-1} and incidence kIm^{k-1} . Since $c \leq \lfloor 2m/5 \rfloor \leq 2m/5$, the degree is an integer at most $2m^k/5$ and so at most $\lfloor 2m^k/5 \rfloor$. For Reimer, the requirement $kIm^{k-1} \geq \lceil km^k \log_2(m)/2 \rceil$ is met as soon as the integer left side dominates the real right side, which cancels to $2I \geq m \log_2 m$, i.e. to the integer inequality

$$m^m \leq 2^{2I},$$

and that is exactly the base condition $I \geq R_m$, because R_m is the least r with $2^{2r} \geq m^m$. *One base-level Reimer check certifies every power.*

Consequently a cloned-coordinate witness carries a valid infinite construction. The 70-row family $\mathcal{C}$ at $n = 8$, whose coordinates 0 and 1 coincide, has $\varepsilon_{\vee}(\mathcal{C}) = 317/490$ and certified $A_+(\mathcal{C}) \leq -3/80$, so its powers give

$$c_{\text{cl}}^*(n) \geq \frac{3}{80} \lfloor n/8 \rfloor, \quad \text{slope } \frac{3}{640} = 0.0046875,$$

against $1/250 = 0.004$ for the seven-coordinate base: an improvement by exactly $75/64$. With Lemma 8 the numerator is pinned between $\frac{3}{640}n - \frac{3}{80}$ and $\frac{4}{5}n$. The powers are audited by direct instantiation; at $k = 2$ the 4,900-row product rebuilt from its rows has defect $210171/240100 = 1 - (173/490)^2$ exactly.

The mechanism is worth isolating. Cloning a coordinate changes neither size, defect, nor join structure, and it raises incidence. For the 70-row bases it buys feasibility: size 70 is impossible at seven coordinates, where $7 \lfloor 2 \cdot 70/5 \rfloor = 196 < R_{70} = 215$. For the fifteen-row base it buys negativity instead: the collapsed five-coordinate core is already admissible but has positive objective, and three copies of one distinguished coordinate make it negative. The objective direction is not universal — cloning any of the other four core coordinates raises A_+ — and the exact order-reweighting law below replaces the earlier false monotonicity claim.

A further complete class, $S_3 \times S_5$, adds 13,470 canonical families of which 54 are float-negative and 52 are certified negative. Its best sits at the maximum admissible size $m = 80$ with incidence $256 = 8 \cdot 32$, and has certified ratio at least $2432/58575 = 0.0415194$, the largest among separating families. That class also shows the float objective failing in both directions: three of its float negatives are exactly non-negative, and one family it reported as non-negative is exactly negative. Every number quoted here is from the exact route.

6.2 Complete block-symmetric coverage at eight coordinates

Since $S_k \times S_{8-k}$ pairs k with $8 - k$, the classes $k \in \{1, 2, 3, 4\}$ exhaust every block-symmetric family on eight coordinates. All four were enumerated — 52,494,332 raw masks, 27,431 canonical admissible families, 150 screen-negative — and every candidate with screen value below 0.01 was re-evaluated exactly, certifying 147.

Two further records come from $S_4 \times S_4$. The fifteen-row family $\mathcal{T}$ with all eight degrees at the cap 6 and incidence $48 \geq R_{15} = 30$ has

$$\varepsilon_{\vee}(\mathcal{T}) = \frac{4}{9} = 0.4444\dots, \quad A_+(\mathcal{T}) \leq -\frac{1}{2100},$$

the lowest certified defect produced directly on eight coordinates, improving $139/245$. It is worth noting what this does *not* require: low defect needed no large family, since Lemma 9's pair bound only forces $m \geq 2.12$ at this defect. What it did require is the dominant set of Corollary 9's consequence, and $\mathcal{T}$ has one — it contains [8], against the threshold $\frac{8}{5}\bar{s} = 128/25$.

The separating 75-row family $\mathcal{S}$, all degrees at the cap 30 and incidence $240 \geq R_{75} = 234$, has $\varepsilon_{\vee}(\mathcal{S}) = 284/375$ and $A_+(\mathcal{S}) \leq -177/5000$, so its powers give slope $177/40000 = 0.004425$ against $1/250 = 0.004$. The growth constant therefore improves by 10.6% without admitting a non-separating base, so that improvement does not depend on the normalization convention; the cloned base retains the overall record $3/640$. The ratio $531/11360 = 0.0467\dots$ is the largest among separating families.

Finally, the class data confirm the diagnosis of the screen: $k = 1$ and $k = 4$ produce no screen error at all, while $k = 2$ and $k = 3$ produce four false negatives and one missed witness between them. Block-pattern orders represent the objective exactly when the family's automorphism group is the block group, and not otherwise.

6.3 All defect-free extensions and the local hard blocker

There is a complete replacement for the earlier informal cloning heuristic. Given $\mathcal{U} \subseteq \mathcal{F}$, append a coordinate whose one-rows are exactly $\mathcal{U}$. Projection onto the old coordinates shows that no failed union can be repaired. If $Z = A \cup B \in \mathcal{F}$, the old success remains a success exactly when

$$\mathbf{1}_{\mathcal{U}}(Z) = \mathbf{1}_{\mathcal{U}}(A) \vee \mathbf{1}_{\mathcal{U}}(B). \quad (22)$$

Thus the defect is preserved precisely for the *join-consistent* row subsets satisfying (22). Equivalently, $\mathcal{U}$ is an up-set inside $(\mathcal{F}, \subseteq)$ and every successful join landing in $\mathcal{U}$ has a part in $\mathcal{U}$. If $|\mathcal{U}| \leq \lfloor 2m/5 \rfloor$, admissibility is automatic: old degrees are fixed, the new degree is $|\mathcal{U}|$, and incidence increases against a fixed R_m .

Global Boolean join-homomorphisms are only constants and ORs of coordinates, but (22) sees only successful joins of this particular family, so its class can be larger. Exhausting all 2^{15} row subsets of the five-coordinate ‘n8tiny’ core finds 57 join-consistent subsets and six usable under the cap: the five old columns and one genuinely new column. Exact evaluation shows that only cloning coordinate 2 improves A_+ ; cloning any of the other four raises it, and the new column raises it to +0.114769081463. Three concentrated copies of coordinate 2 make the objective negative, whereas spreading the same three copies over distinct coordinates leaves it +0.006494611034.

The general cloning law is sharper. In an order of a family with arbitrary clone multiplicities, keep the first copy in each class and delete the rest. Later copies are deterministic, contribute zero entropy, and leave every row fiber unchanged. If coordinate i has multiplicity r_i , the collapsed original order has the exact exponential-race (Plackett–Luce) law

$$\Pr_{\mathbf{r}}(\pi) = \prod_{k=1}^n \frac{r_{\pi_k}}{\sum_{j=k}^n r_{\pi_j}}, \quad A_+(\tilde{\mathcal{F}}_{\mathbf{r}}) = \sum_{\pi \in S_n} \Pr_{\mathbf{r}}(\pi) A_{+, \pi}(\mathcal{F}). \quad (23)$$

For any target order, choosing $r_{\pi_k} = R^{n-k}$ gives it probability at least $(1-1/R)^{n-1} \geq 1-(n-1)/R$. Hence

$$\inf_{\mathbf{r}} A_+(\tilde{\mathcal{F}}_{\mathbf{r}}) = \min_{\pi} A_{+, \pi}(\mathcal{F}), \quad \sup_{\mathbf{r}} A_+(\tilde{\mathcal{F}}_{\mathbf{r}}) = \max_{\pi} A_{+, \pi}(\mathcal{F}). \quad (24)$$

The extrema need not be attained at finite multiplicity, but sign reachability is finite: some clone multiset is negative if and only if some original fixed order is negative. For one cloned coordinate, the earlier rank law is

$$\Pr_r[j] = \frac{\binom{n+r-j-2}{r-1}}{\binom{n+r-1}{r}},$$

whose mass outside rank zero is $(n-1)/(n+r-1)$. Thus convergence is $O(1/r)$, not geometric; for the record core, the coordinate-2 limit is $-0.002338401508\dots$

The $n = 9, 10$ local search covered 13 and 15 selected sizes respectively, including the admissible maxima 145 and 255; every added size at least 100 had defect above 0.59. It found a normalized core

$$\mathcal{R} = (0, 2, 4, 6, 8, 9, 16, 22, 31, 32, 96, 105, 112, 125, 127) \subseteq 2^{[7]}$$

with degrees $(5, 5, 6, 6, 6, 6, 5)$, incidence $39 \geq R_{15} = 30$, the full set, and defect $14/45 < 2/5$. It is not negative:

$$A_+(\mathcal{R}) \in [0.305904110097, 0.305904110098], \quad \min_{\pi} A_{+, \pi}(\mathcal{R}) \geq 0.145914214376.$$

Its join-consistent class has seven old columns and exactly one new usable column. Adding that column leaves every one of the resulting 40320 fixed orders positive, with minimum at least

0.122963869853. Since the successful join relation is invariant under a defect-free extension, these two convex-hull certificates cover every finite extension sequence: the sub-2/5 family cannot be rescued by any defect-free coordinate mechanism.

A separate Q -minimizing walk found one $n = 9, m = 20$ family at defect 19/50 passing the necessary condition $(1 - \alpha)Q - \log_2 m \leq 0$. Its Q -only room was -0.003072867378 , but sampled $C_+ \approx 4.82$ contributed $+0.171684\dots$, moving sampled A_+ to $+0.168611231079$. Thus C_+ , not Q , is binding on this survivor. Removing its one duplicate column gives a normalized eight-coordinate core with

$$A_+ \in [0.172732849547, 0.172732849548], \quad \min_{\pi} A_{+, \pi} \geq 0.042394032883,$$

so every clone configuration is again excluded. A sampled-order full-average walk remained at $A_+ \geq 0.128833928728$ on its reported frontier; this is discovery evidence only. It nevertheless identified the scoring error: full-order Q is only a rejection gate, and a full average hides the negative fixed orders that Proposition 25 can amplify.

A subsequent minimum-fixed-order walk succeeded. After removing its one duplicate column, its normalized core is

$$\mathcal{V} = (0, 1, 2, 4, 5, 8, 10, 43, 64, 190, 192, 193, 245, 254, 255) \subseteq 2^{[8]}.$$

It has degrees $(6, 6, 6, 6, 4, 5, 6, 6)$, incidence $45 \geq R_{15} = 30$, contains $[8]$, and has defect 14/45. Its uniform order average is positive, $A_+(\mathcal{V}) \in [0.230029313333, 0.230029313334]$, but the exact rational enclosure at order $(6, 1, 2, 0, 3, 4, 5, 7)$ lies in

$$[-0.003540262562, -0.003540262561].$$

Using the geometric clone ratio $R = 41$, equivalently multiplicities

$$(2825761, 4750104241, 115856201, 68921, 1681, 41, 194754273881, 1),$$

and reweighting all 40320 exact fixed-order enclosures gives a 199623130728-coordinate family with the same 15 rows and defect, incidence 1197738780965, and

$$A_+ \in [-0.000084288238, -0.000084288237] < 0.$$

Thus the certified negative defect in the displayed cap/Reimer class is now

$$\boxed{14/45 = 0.3111\dots < 2/5},$$

not 4/9. The core is normalized, but the cloned witness has repeated columns; the separating negative frontier remains 1144/1875. The exact two-sided certificate is `certificates/gate_b_subtwofifths_clone_rational_v1.json`.

Finally, the natural cap-only floor is false. Chase–Lovett [2] construct

$$\mathcal{F}_n = \{A : |A| = \psi n + n^{2/3}\} \cup \{A : |A| \geq (1 - \psi)n\}, \quad \psi = \frac{3 - \sqrt{5}}{2},$$

which has $\varepsilon_V = o(1)$ and every frequency at most $\psi + o(1) < 2/5$. For large n it is active and separating and contains $[n]$, so neither normalization nor a dominant set yields a positive floor. It is excluded here only by Reimer:

$$\frac{\bar{s}}{n} \rightarrow \psi, \quad \frac{\log_2 |\mathcal{F}_n|}{2n} \rightarrow \frac{h(\psi)}{2}, \quad \frac{h(\psi)}{2} - \psi > 0.0977433528.$$

Thus any floor for the actual Gate B class must use Reimer essentially.

The role of Reimer has two exact limits. First, at zero defect it is automatic: Reimer’s average-set-size theorem [1] applies to every union-closed family. Such a family contains $T = \bigcup_{A \in \mathcal{F}} A$; on its active support, the cap gives $\bar{s} \leq 2|T|/5$, so T is automatically stronger than the dominant set required above. Deleting zero and duplicate columns preserves row identities, union closure, and the cap, so normalization adds no escape. A positive floor reaching zero would therefore prove the $2/5$ frequency theorem under either convention.

Second, Reimer cannot be recovered continuously from pair defect. With $\Delta_R = \frac{1}{2} \log_2 |\mathcal{F}| - \bar{s}$, the same Chase–Lovett sequence has

$$\frac{\Delta_R(\mathcal{F}_n)}{n} \longrightarrow \frac{h(\psi)}{2} - \psi > 0.0977433528 \quad \text{while} \quad \varepsilon_V(\mathcal{F}_n) \longrightarrow 0.$$

Thus no inequality $\Delta_R \leq n g(\varepsilon_V)$ with $g(t) \rightarrow 0$ follows even after imposing the cap, normalization, and a full set. Any successful positive-defect argument must use Reimer as a hard hypothesis or add structure beyond the ordered-pair defect. Cambie’s two-strategy ceiling $c^* \approx 0.3823455333667027$ [3] is a closely related obstruction, but its matching lower verification is numerical/graphical in the cited source; the exact two-atom upper obstruction is all that is used here.

7 Scope and limitations

Every Cartesian-power sequence used for the unrestricted theorem satisfies $\varepsilon_V \rightarrow 1$. This is no limitation for the supremum (5), but it is why that theorem does not settle the distinct local question $\varepsilon_V \rightarrow 0$. The restriction is structural: closure success multiplies, so

$$\varepsilon_V(\mathcal{F} \boxtimes \mathcal{G}) \geq \max\{\varepsilon_V(\mathcal{F}), \varepsilon_V(\mathcal{G})\},$$

and no power can dilute the defect. Failures also occur in off-diagonal pairs, so $\varepsilon_V \geq 2/m^2$ for a non-union-closed family and $-A_+/\varepsilon_V \leq m^2 \log_2(m)/2$.

The exact local conclusions have deliberately different scopes. The least certified negative defects are $14/45$ in the displayed cap/Reimer class and $1144/1875$ when active separation is also required. The first normalized $14/45$ core $\mathcal{R}$ and every defect-free extension of it are certified positive; the row-changing search found a different normalized core $\mathcal{V}$ with a negative fixed order, and the exact clone law turns that order into the certified non-separating witness above. Sampling selected the core and order but proves nothing: the final fixed-order and cloned averages are two-sided rational enclosures. The defect remains the fixed value $14/45$, so this still does not determine the limit c_{loc} as $\varepsilon_V \rightarrow 0$.

Proposition 22 also explains why a theorem from cap and dominant-set data alone cannot close the local problem: the Chase–Lovett construction refutes exactly that statement under both normalization conventions. Reimer excludes the construction, leaving a sharply stated open problem rather than a plausible but false floor. At the endpoint, a zero-defect admissible family would be union-closed with every frequency at most $2/5$, so a universal positive Gate B floor would prove a $2/5$ frequency theorem. This paper does not claim that result.

The finite negative bases are independently checked by 256-bit Arb, standard-library dyadic intervals, and exact rational arithmetic where those routes apply. The exact rational route uses the true action interval and no interval library; all $n \leq 7$ orders are evaluated directly, while the eight-coordinate certificates use exact automorphism orbits whose equal-size property and group action are tested. The tensorization lemmas and the extension/convex-hull arguments remain human mathematics with exhaustive finite falsification controls. None of the certificates is a proof-assistant artifact, and formalization remains open.

References

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